```
import numpy as np
```

```
#1.create a numpy arraycontaining 5 zeros using'np.zeros'
a=np.zeros(5)
a
```

```
array([0., 0., 0., 0., 0.])
```

```
#2.create a 3*3 matrix filled with ones using 'np.ones'
a= np.ones((3,3))
a
```

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

```
#3.create a 4*4 matrix  using 'np.eye'
a= np.eye(4,4)
a
```

```
array([[1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.]])
```

```
#4.create a digonal matrix using the value [10,20,30] with 'np.diag
a=np.array([10,20,30])
a
b=np.diag(a)
b
```

```
array([[10,  0,  0],
       [ 0, 20,  0],
       [ 0,  0, 30]])
```

```
#5.genrate 8 random integer b/w 1,50 using'np.random.randint'
a=np.random.randint(1,50,8)
a
```

```
array([19, 34, 17, 37,  9, 20, 29, 10])
```

```
#6.genrate a 2*3 array of random decimal value b/w 0,1 using'np.random.rand'
a=np.random.rand(6).reshape(2,3)
a=np.random.rand(2,3)   # dono same kaam krenge
a
```

```
array([[ 1.18727565, -0.14295466, -0.30479716],
       [-0.94212439, -1.17427494,  0.25289346]])
```

```
#7.generate num from 5,50 with a step size of 5 using'np.arange'
a=np.arange(5,51,5)
a
```

```
array([ 5, 10, 15, 20, 25, 30, 35, 40, 45, 50])
```

```
#8.generate 10 equally spaced num b/w 0,100 using' np.linspace'
a=np.linspace(0,100,10)
a
```

```
array([  0.        ,  11.11111111,  22.22222222,  33.33333333,
        44.44444444,  55.55555556,  66.66666667,  77.77777778,
        88.88888889, 100.        ])
```

```
#9.combine two numpy arrays [1,2,3] and [4,5,6] horizontally using'np.hstack'
a=np.array([1,2,3])
b=np.array([4,5,6])
np.hstack((a,b))
```

```
array([1, 2, 3, 4, 5, 6])
```

```
#10.combine two numpy arrays [10,20,30] and [40,50,60] vertically using'np.vstack'
a=np.array([10,20,30])
b=np.array([40,50,60])
np.vstack((a,b))
```

```
array([[10, 20, 30],
       [40, 50, 60]])
```

```
#11.create a 2*4 matrix filled with zeros using'np.zeros'
a=np.zeros((2,4))
a
```

```
array([[0., 0., 0., 0.],
       [0., 0., 0., 0.]])
```

```
#12.extract the digonal elements from the following matrix using'np.diag'
#arr=np.array([[1,2,3],[4,5,6],[7,8,9]])
arr=np.random.randint(1,10,size=(3,3))
arr
a=np.array([4,7,9])
arr=np.diag(a)
arr
```

```
array([[4, 0, 0],
       [0, 7, 0],
       [0, 0, 9]])
```

```
#13.create two 2*2 numpy matrix and combine them horizontally using'np.hstack'and vertically using'np.vstack'
a=np.array([[1,2],
   [3,4]])
b=np.array([[5,6],
   [7,8]])
np.hstack((a,b))
np.vstack((a,b))
```

```
array([[1, 2],
       [3, 4],
       [5, 6],
       [7, 8]])
```

```
Start coding or generate with AI.
```